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(54) INFORMATION PROCESSING SYSTEM AND NON-TRANSITORY COMPUTER READABLE MEDIUM

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(57) **ABSTRACT**

An information processing system includes a processor configured to, when a solution request for a failure of a device in which the failure has occurred is received from a user who uses the device, request the user to provide information related to solution for the failure, and display, to the user, an indication related to an amount of work required from the user when a transition is made to another way of solution for the failure.

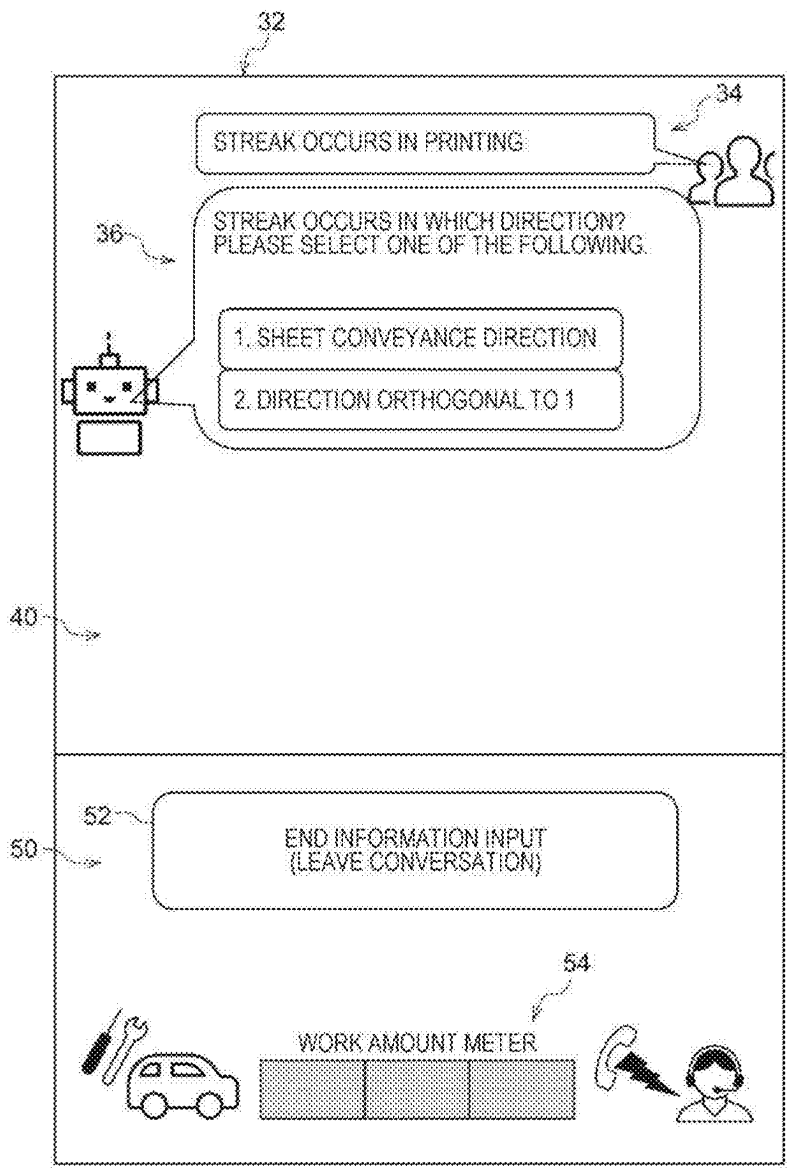


FIG. 1

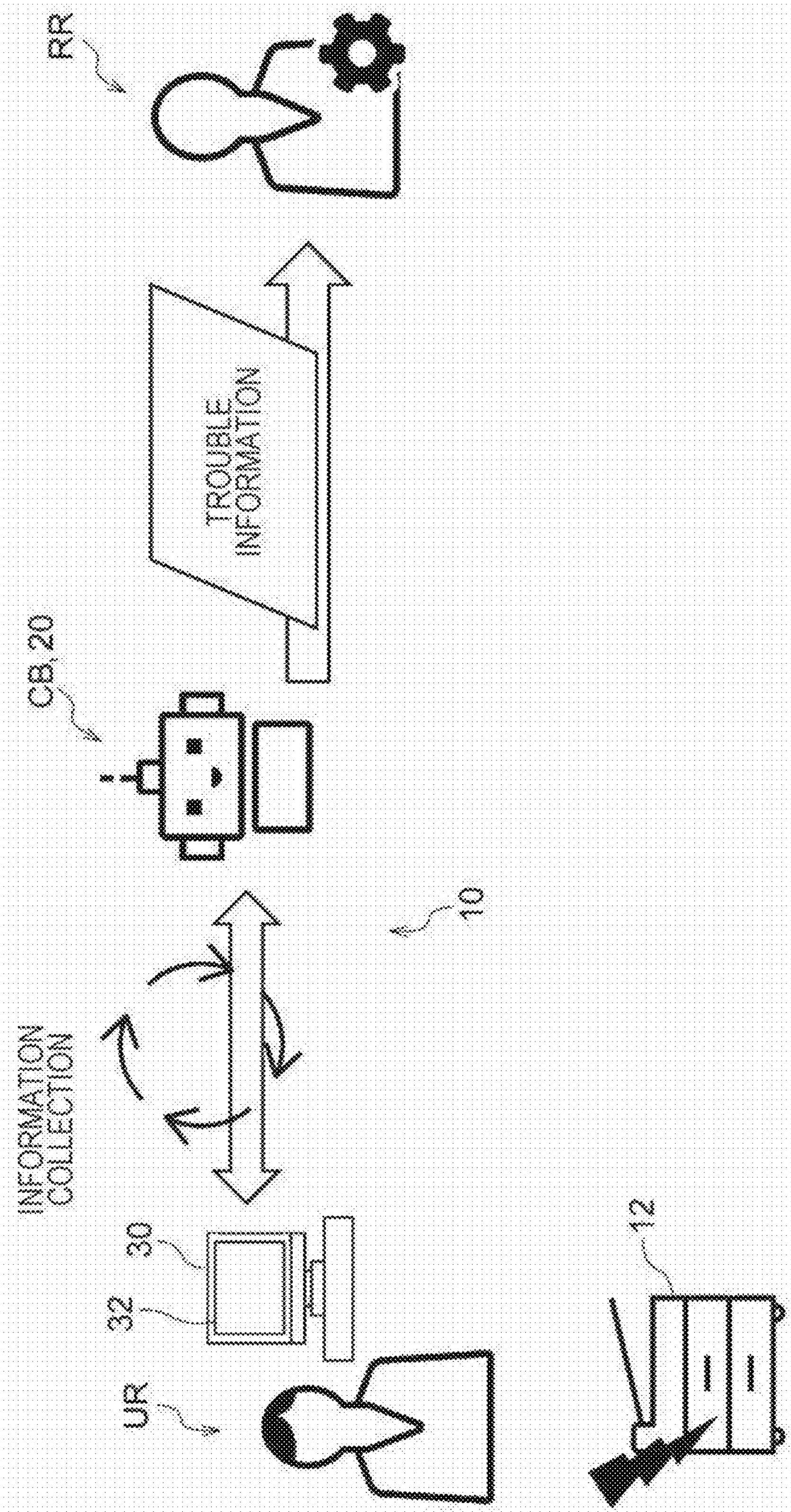

RR
TROUBLE INFORMATION
CB, 20
10
INFORMATION COLLECTION
30
32
UR
12

FIG. 2

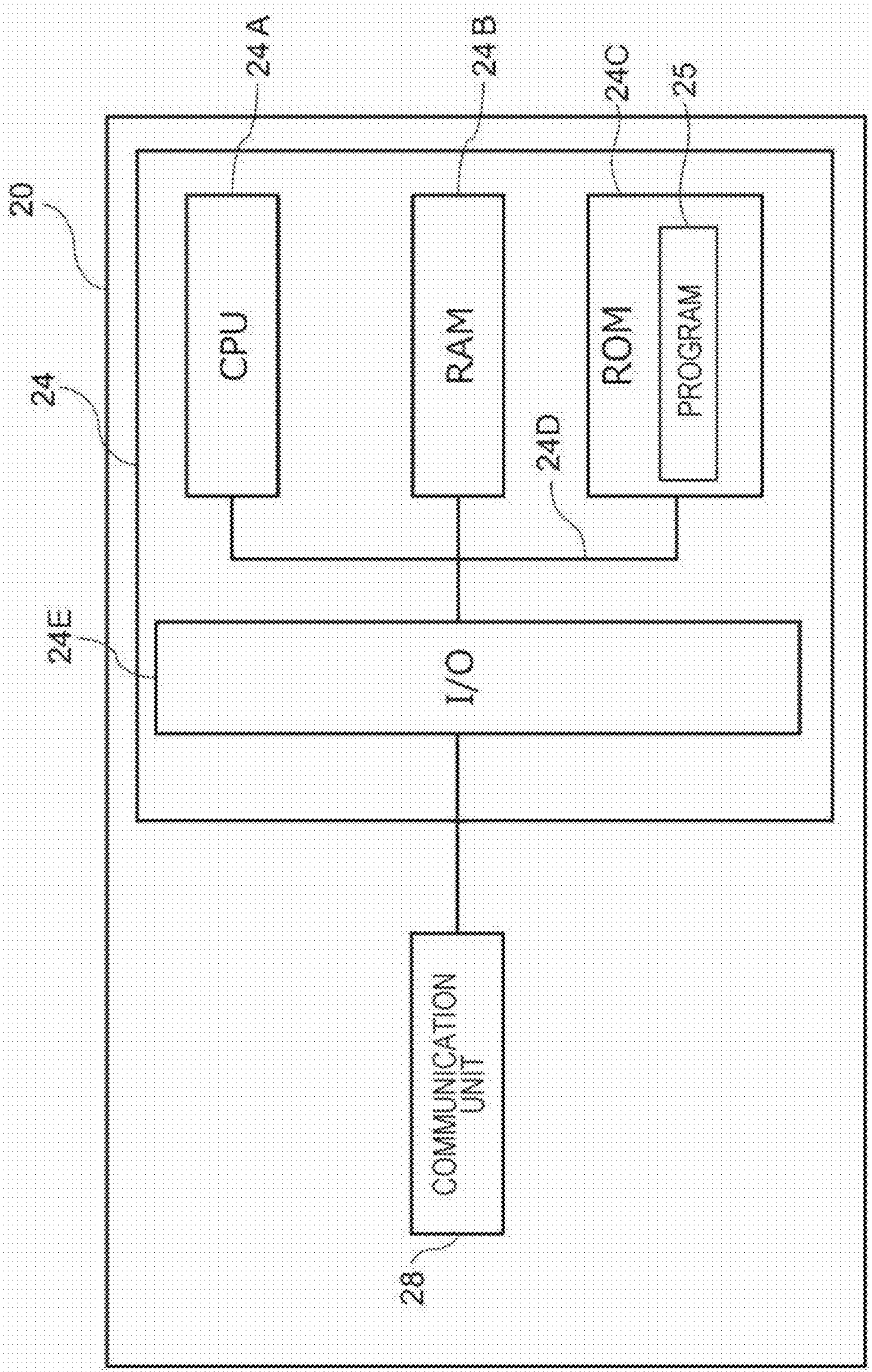

20
24
24 A
CPU
24 B
RAM
24C
ROM
25
PROGRAM
24D
24E
I/O
COMMUNICATION UNIT
28

FIG. 3

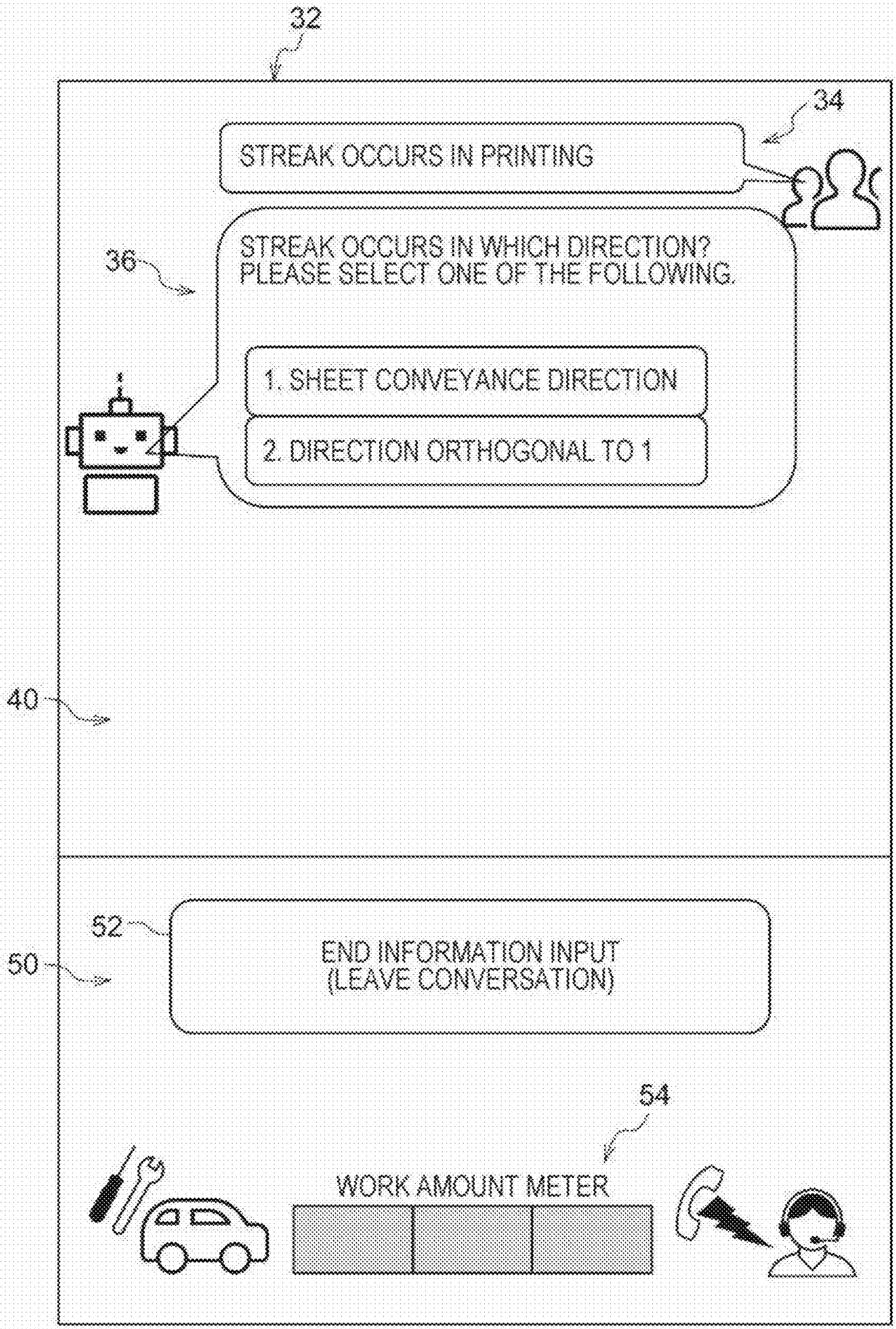

32
34
STREAK OCCURS IN PRINTING
36
STREAK OCCURS IN WHICH DIRECTION?
PLEASE SELECT ONE OF THE FOLLOWING.
1. SHEET CONVEYANCE DIRECTION
2. DIRECTION ORTHOGONAL TO 1
40
52
END INFORMATION INPUT
(LEAVE CONVERSATION)
50
54
WORK AMOUNT METER

FIG. 4

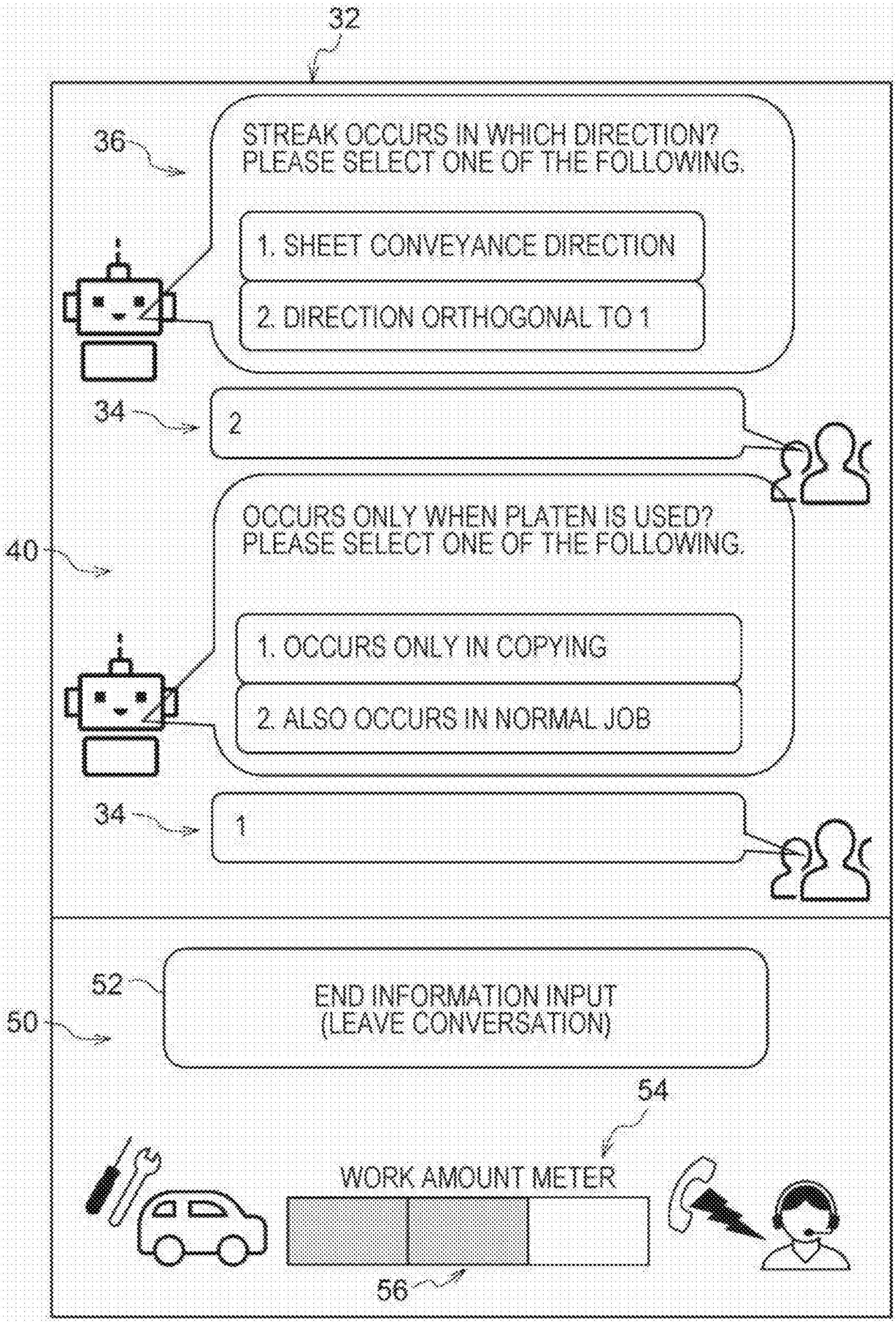

32
36
STREAK OCCURS IN WHICH DIRECTION?
PLEASE SELECT ONE OF THE FOLLOWING.
1. SHEET CONVEYANCE DIRECTION
2. DIRECTION ORTHOGONAL TO 1
34
2
40
OCCURS ONLY WHEN PLATEN IS USED?
PLEASE SELECT ONE OF THE FOLLOWING.
1. OCCURS ONLY IN COPYING
2. ALSO OCCURS IN NORMAL JOB
34
1
52
50
END INFORMATION INPUT
(LEAVE CONVERSATION)
54
WORK AMOUNT METER
56

FIG. 5

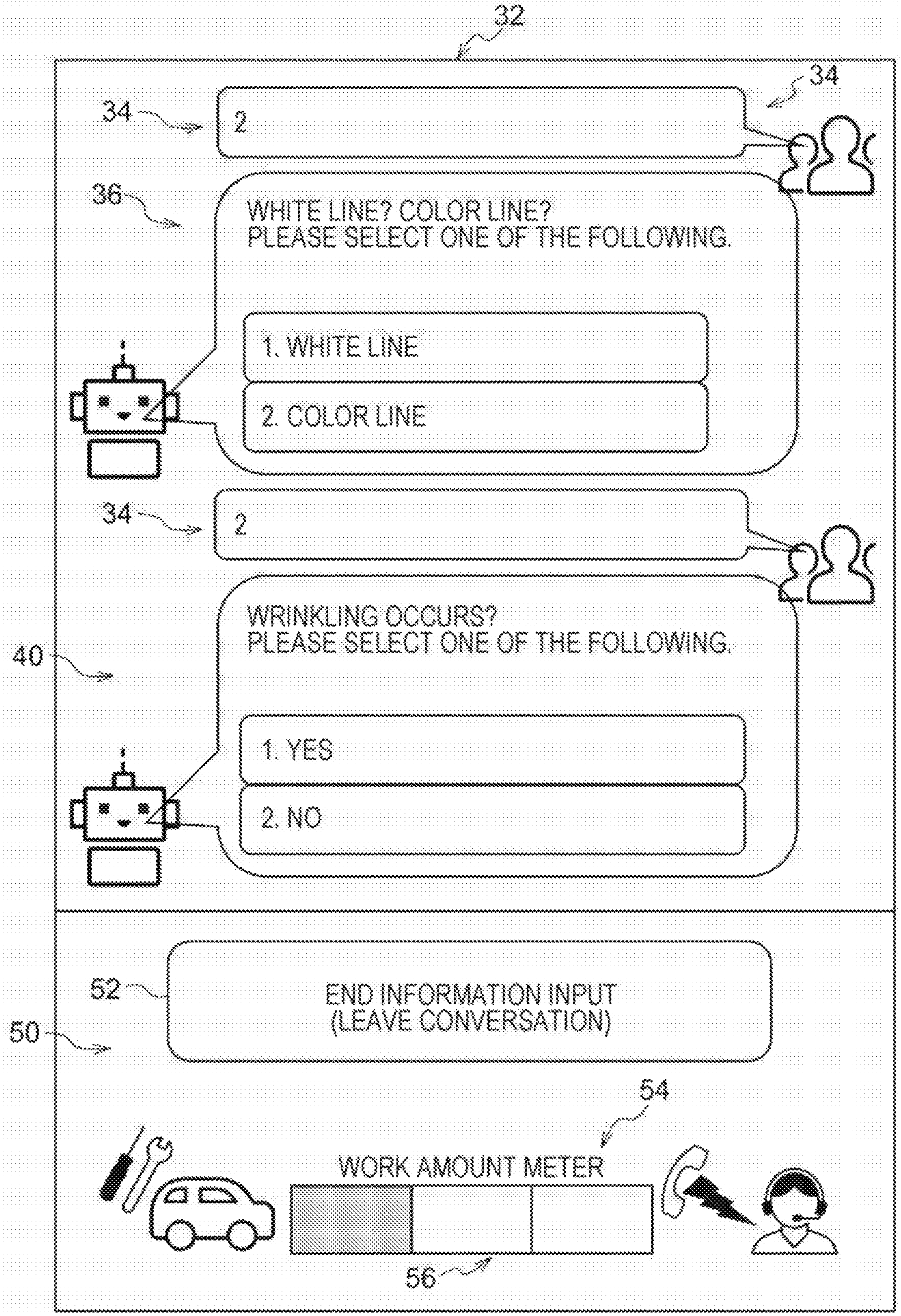

32
34
34
2
36
WHITE LINE? COLOR LINE?
PLEASE SELECT ONE OF THE FOLLOWING.
1. WHITE LINE
2. COLOR LINE
34
2
40
WRINKLING OCCURS?
PLEASE SELECT ONE OF THE FOLLOWING.
1. YES
2. NO
52
END INFORMATION INPUT
(LEAVE CONVERSATION)
50
54
WORK AMOUNT METER
56

FIG. 6

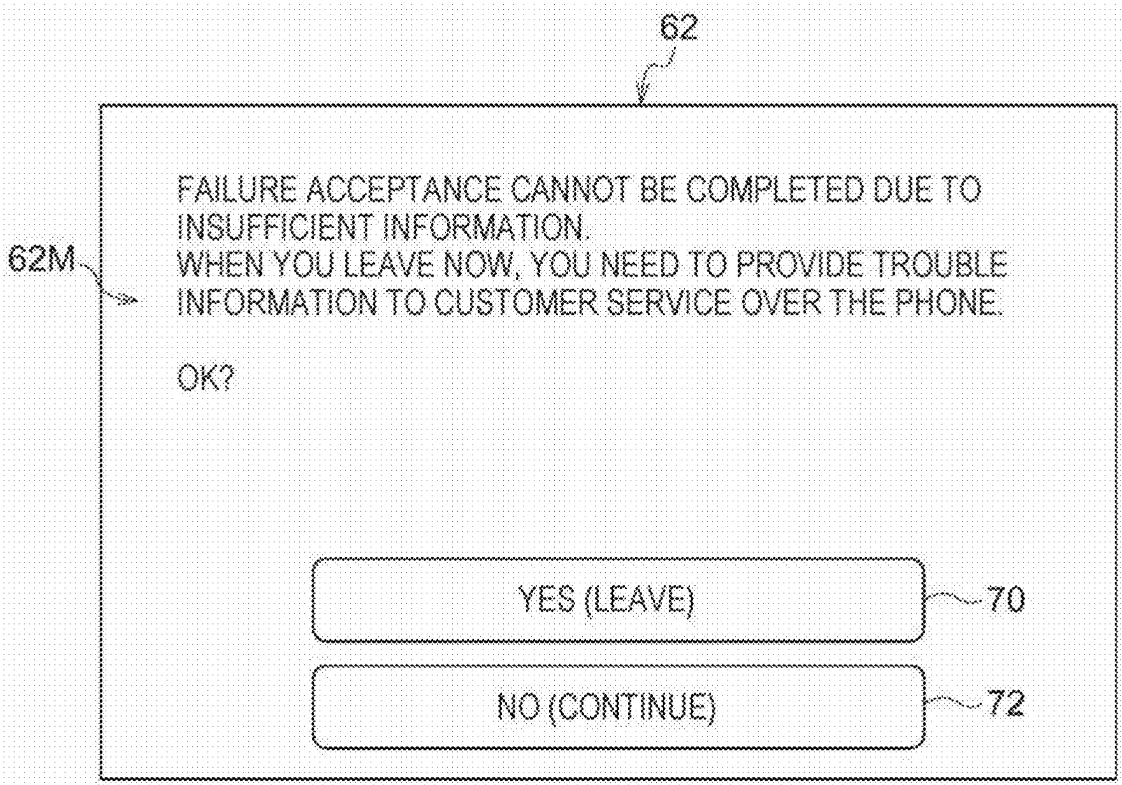

62
62M
FAILURE ACCEPTANCE CANNOT BE COMPLETED DUE TO INSUFFICIENT INFORMATION.
WHEN YOU LEAVE NOW, YOU NEED TO PROVIDE TROUBLE INFORMATION TO CUSTOMER SERVICE OVER THE PHONE.
OK?
YES (LEAVE)
70
NO (CONTINUE)
72

FIG. 7

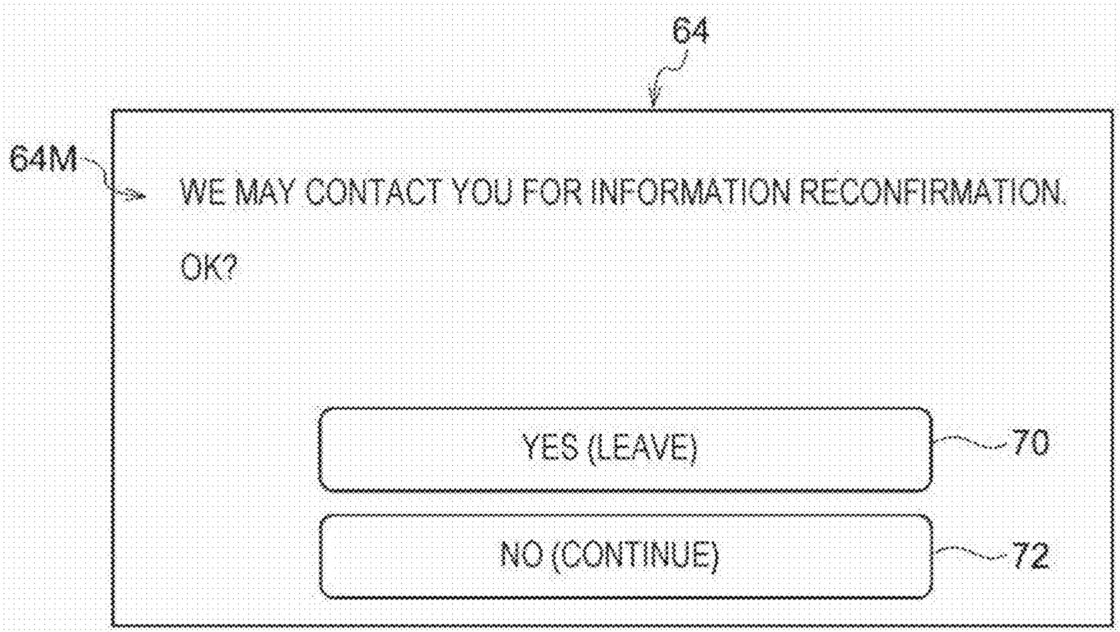

64
64M
WE MAY CONTACT YOU FOR INFORMATION RECONFIRMATION.
OK?
YES (LEAVE)
70
NO (CONTINUE)
72

FIG. 8

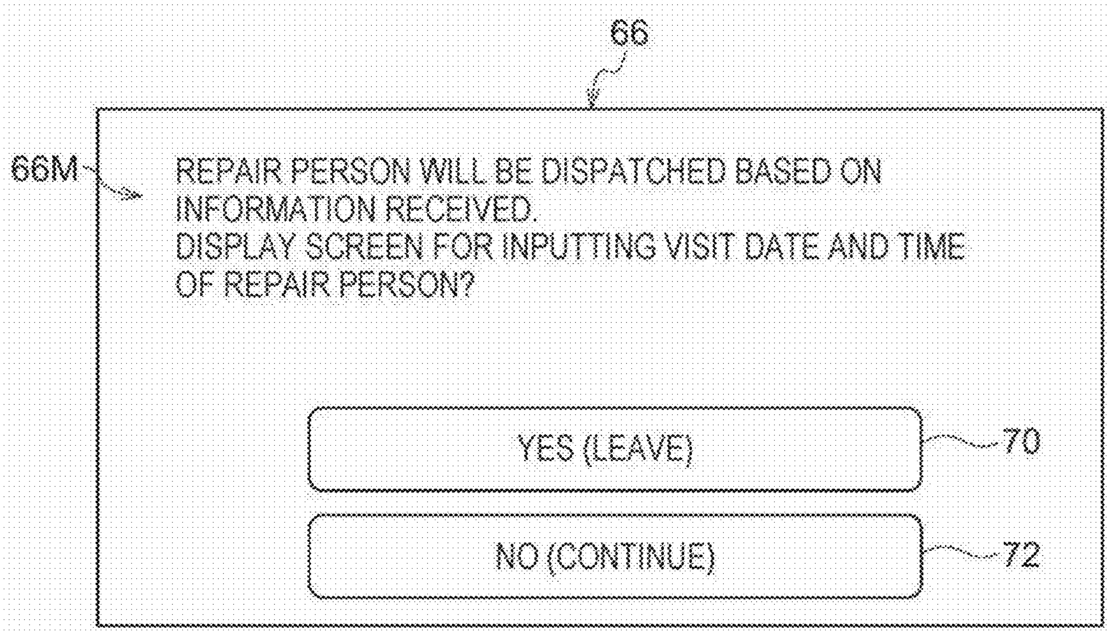
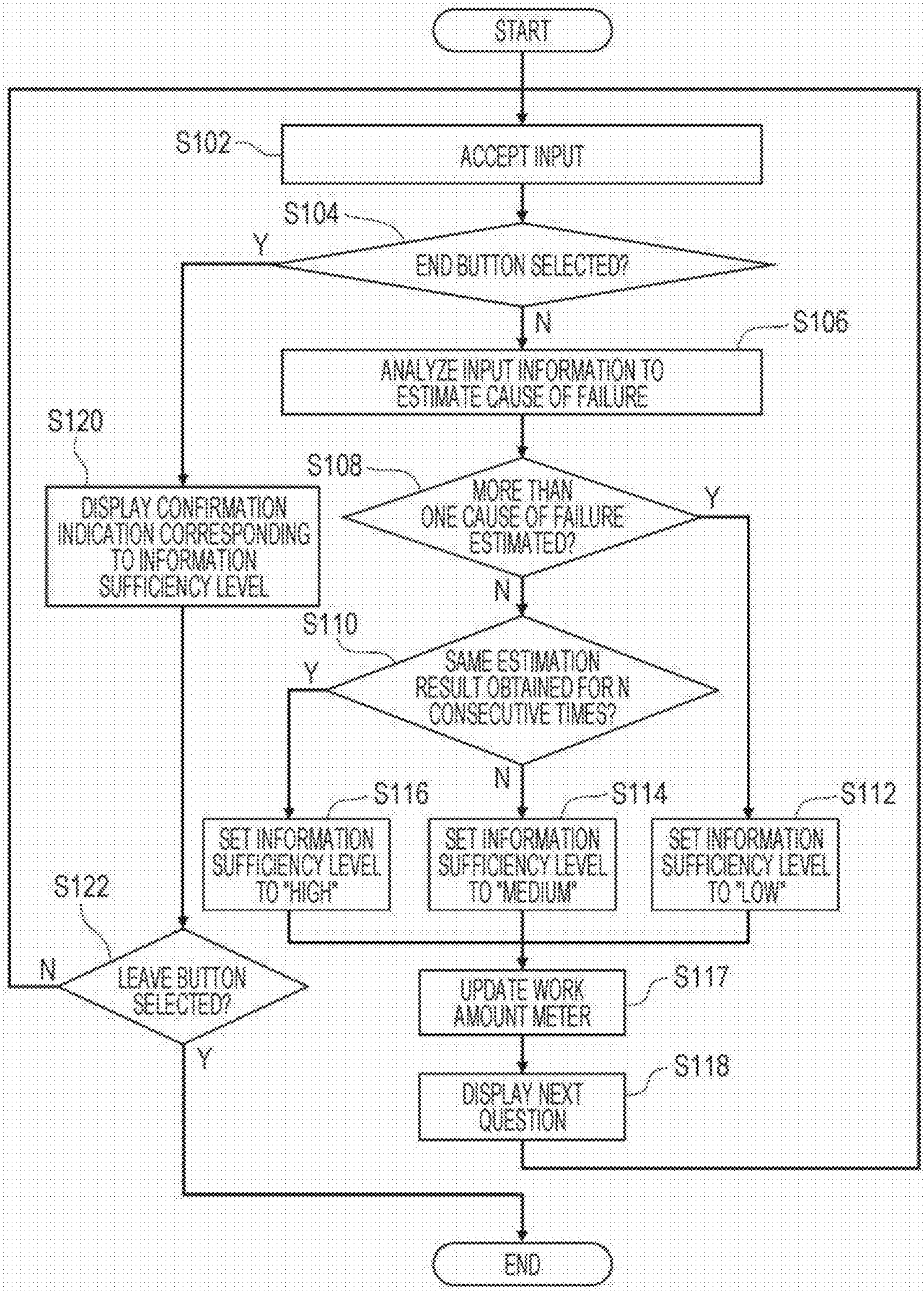

66
66M
REPAIR PERSON WILL BE DISPATCHED BASED ON INFORMATION RECEIVED.
DISPLAY SCREEN FOR INPUTTING VISIT DATE AND TIME OF REPAIR PERSON?
YES (LEAVE)
70
NO (CONTINUE)
72

FIG. 9


START
S102
ACCEPT INPUT
S104
END BUTTON SELECTED?
Y
N
S106
ANALYZE INPUT INFORMATION TO ESTIMATE CAUSE OF FAILURE
S108
MORE THAN ONE CAUSE OF FAILURE ESTIMATED?
Y
N
S110
SAME ESTIMATION RESULT OBTAINED FOR N CONSECUTIVE TIMES?
Y
N
S116
SET INFORMATION SUFFICIENCY LEVEL TO "HIGH"
S114
SET INFORMATION SUFFICIENCY LEVEL TO "MEDIUM"
S112
SET INFORMATION SUFFICIENCY LEVEL TO "LOW"
S117
UPDATE WORK AMOUNT METER
S118
DISPLAY NEXT QUESTION
S120
DISPLAY CONFIRMATION INDICATION CORRESPONDING TO INFORMATION SUFFICIENCY LEVEL
S122
LEAVE BUTTON SELECTED?
N
Y
END

INFORMATION PROCESSING SYSTEM AND NON-TRANSITORY COMPUTER READABLE MEDIUM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-052468 filed Mar. 27, 2024.

BACKGROUND

(i) Technical Field

[0002] The present disclosure relates to an information processing system and a non-transitory computer readable medium.

(ii) Related Art

[0003] For example, in International Publication No. 2020/240838, an instruction to start a conversation with a chatbot and attribute information of a user acquired by an information processing terminal used by the user are received from the information processing terminal. There has been disclosed an information processing apparatus that starts a conversation with a chatbot in an information processing terminal using script information determined based on attribute information of a user by referring to a storage unit that stores script information corresponding to each of a plurality of user attributes.

SUMMARY

[0004] Aspects of non-limiting embodiments of the present disclosure relate to an information processing system that collects information provided by a user who uses a device in which a failure has occurred, and prompts the user to continue providing information.

[0005] Aspects of certain non-limiting embodiments of the present disclosure address the above advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.

[0006] According to an aspect of the present disclosure, there is provided an information processing system including a processor configured to, when a solution request for a failure of a device in which the failure has occurred is received from a user who uses the device, request the user to provide information related to solution for the failure, and display, to the user, an indication related to an amount of work required from the user when a transition is made to another way of solution for the failure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Exemplary embodiments of the present disclosure will be described in detail based on the following figures, wherein:

[0008] FIG. **1** is a diagram illustrating a response system according to an exemplary embodiment of the present disclosure;

[0009] FIG. **2** is a diagram illustrating a system configuration of a chatbot according to the exemplary embodiment of the present disclosure;

[0010] FIG. **3** is a diagram illustrating contents displayed on an input screen operated by a user according to the exemplary embodiment of the present disclosure;

[0011] FIG. **4** is a diagram illustrating contents displayed on a display unit according to the exemplary embodiment of the present disclosure following FIG. **3**, and is a diagram illustrating the contents in a state where information is provided from the user and a new question is displayed;

[0012] FIG. **5** is a diagram illustrating contents displayed on the display unit according to the exemplary embodiment of the present disclosure following FIG. **4**, and is a diagram illustrating the contents in a state where the user has continued providing information;

[0013] FIG. **6** is a diagram illustrating a first confirmation indication that is an example of an indication related to work details according to the exemplary embodiment of the present disclosure and that is displayed on the input screen operated by the user in a case where the user attempts to leave in a state with a low information sufficiency level;

[0014] FIG. **7** is a diagram illustrating a second confirmation indication that is an example of an indication related to work details according to the exemplary embodiment of the present disclosure and that is displayed on the input screen operated by the user in a case where the user attempts to leave in a state with a medium information sufficiency level;

[0015] FIG. **8** is a diagram illustrating a third confirmation indication that is an example of an indication related to work details according to the exemplary embodiment of the present disclosure and that is displayed on the input screen operated by the user in a case where the user attempts to leave in a state with a high information sufficiency level; and

[0016] FIG. **9** is a flowchart illustrating an operation procedure of the response system according to the exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

[0017] Hereinafter, an example of an exemplary embodiment of the present disclosure will be described with reference to the drawings. Note that in the drawings, the same or equivalent components and parts are denoted by the same reference signs. In addition, dimensional ratios in the drawings are exaggerated for convenience of description and may be different from actual ratios.

Exemplary Embodiment

[0018] FIG. **1** is a diagram illustrating a usage example of a response system **10**, which is an example of an information processing system according to the present exemplary embodiment. As illustrated in FIG. **1**, the response system **10** includes an input device **30** for a user UR and a computer **20** to which the input device **30** is connected via a network, and is used when the user UR who is a person using a printer **12** makes an inquiry about the printer **12**.

(Computer **20**)

[0019] As illustrated in FIG. **2**, the computer **20** includes a communication unit **28** and a control unit **24**. Furthermore, as illustrated in FIG. **2**, the control unit **24** includes a CPU

24A, a RAM 24B, a ROM 24C, and an input/output interface (I/O) 24E. These components are connected to each other via a control bus 24D.

[0020] The communication unit **28** is a component for communicating with other devices via a network (not illustrated). Specifically, the communication unit **28** communicates with other devices by using wired or wireless communications, as well as communications using the Internet, an intranet, and a public line such as a telephone line.

[0021] The CPU **24**A is a central processing unit, which is an example of a "processor" that executes various programs and controls each unit. The RAM **24**B, as a work area, temporarily stores therein a program **25** or data. The ROM **24**C stores various types of data including a "program" according to the present disclosure.

[0022] In the control unit **24**, the CPU **24**A reads various programs including the processing program **25** from the ROM **24**C, and executes the program **25** using the RAM **24**B as a work area. By executing the processing program **25**, the CPU **24**A realizes various functions for controlling each unit of the computer **20**.

[0023] The computer **20** according to the present exemplary embodiment executes the response system **10** when the CPU **24**A executes the program **25**. The response system **10** is, for example, a so-called chatbot that displays a predetermined question to get an answer from the user UR. Further, the response system **10** prompts the user UR to provide information on a failure of the printer **12** by displaying an input screen **32** for inputting a question **36** on the display screen of the input device **30** for the user UR.

[0024] More specifically, as illustrated in FIG. **1**, when a failure has occurred in the printer **12** used by the user UR, the user UR uses the response system **10**, which is made available on a website of the distributor, by using the input device **30**. Then, the response system **10** provides a repair person RR with the contents of the solution to the failure obtained based on the information provided from the user UR. Thereafter, the repair person RR visits the user UR and addresses the cause of the failure of the printer **12** based on the contents of the solution to the failure provided from the user UR.

[0025] In other words, the use of the response system **10** by the user UR is an example of issuing a solution request for a device failure according to the present exemplary embodiment. Displaying, by the CPU **24**A, the input screen **32** for inputting an answer to the question **36** on the display screen of the input device **30** used by the user UR is an example of receiving a solution request according to the present exemplary embodiment.

[0026] Next, an operation of the response system **10** according to the present exemplary embodiment will be described with reference to FIGS. **3** to **9**.

(Chat Display Area **40** and Status Display Area **50**)

[0027] As illustrated in FIG. **3**, an input screen **32** including a chat display area **40** and a status display area **50** is displayed on the display screen of the input device **30** for the user UR who uses the response system **10** according to the present exemplary embodiment.

[0028] As illustrated in FIG. **3**, the chat display area **40** is an area in which the user UR inputs the status of the failure occurring in the printer **12**. The user UR provides a chatbot CB, which is the response system **10**, with information related to the failure having occurred in the printer **12** by inputting the details, status, and information of the failure occurring in the printer **12**. In addition, in the chat display area **40**, the question **36** with which the chatbot CB requires for further information related to the printer **12** is displayed based on input information **34** related to the failure having occurred in the printer **12**, which has been input by the user UR. The user UR can provide information related to the failure having occurred in the printer **12** as additional input information **34** by answering the question **36** presented by the chatbot CB.

[0029] As illustrated in FIG. **3**, the status display area **50** is an area in which an indication of the amount of work required from the user when a transition is made to another way of solution is displayed in accordance with the sufficiency level of the information provided from the user UR, and a work amount meter **54** that displays the amount of work required from the user is displayed. The work amount meter **54** is, for example, a so-called progress bar that displays, when the user UR inputs the input information **34**, whether the amount of work required from the user is large or small when a transition is made to another way of solution in accordance with the sufficiency level of the information provided from the user UR included in the input information **34**.

[0030] Further, in the status display area **50**, as illustrated in FIG. **3**, an end button **52** used by the user UR for inputting selection to end the conversation with the chatbot CB (to leave the conversation) is displayed.

[0031] In the following description, the "sufficiency level of information" indicates, for example, how precisely the cause of the failure having occurred in the printer **12**, which is estimated based on the input information **34** that is the information provided from the user UR, is narrowed down. Specifically, a higher sufficiency level of information is assumed to indicate that the cause of the failure having occurred in the printer **12** is more precisely narrowed down, and a lower sufficiency level of information is assumed to indicate that the cause of the failure having occurred in the printer **12** is less precisely narrowed down. The amount of work required from the user when a transition is made to another way of solution in accordance with the sufficiency level of information is displayed as a current value **56** of the work amount meter **54**.

[0032] Then, as illustrated in FIGS. **3** to **5**, the user UR in the present exemplary embodiment provides information on the failure occurring in the printer **12** by appropriately inputting the input information **34** in response to the question presented from the chatbot CB displayed in the chat display area **40**.

[0033] As illustrated in FIGS. **4** and **5**, in the response system **10** according to the present exemplary embodiment, each time the user UR answers a question presented from the chatbot CB, a new question is presented from the chatbot CB displayed by the CPU **24**A. As illustrated in FIGS. **4** and **5**, the CPU **24**A narrows down the cause of the failure occurring in the printer **12** based on the input information **34** provided from the user UR every time the user UR answers the question **36** presented from the chatbot CB.

[0034] Then, the CPU **24**A changes the display of the work amount meter **54** based on how precisely the cause of the failure occurring in the printer **12**, which is estimated based on the information provided from the user UR, is narrowed down. For example, in the example illustrated in FIGS. **3** to **5**, in a procedure for receiving the details of the failure from the user UR, which will be described later, the current value **56** of the work amount meter **54** is displayed such that the smaller the number of possible failure cause candidates is, the higher the sufficiency level of information is. More specifically, in the example illustrated in FIGS. **3** to **5**, the progress bar displayed indicating the current value **56** of the amount of work required from the user when a transition is made to another way of solution is shortened in response to an increase in the sufficiency level of the information.

[0035] In addition, in the response system **10** according to the present exemplary embodiment, the user UR can select the end button **52** at any timing. In other words, upon determining that he or she has finished answering the question **36** presented from the chatbot CB, the user UR can leave the conversation with the chatbot CB by intentionally selecting the end button **52**. In the present exemplary embodiment, the user UR pressing the end button **52** to leave the conversation is an example of "attempt to interrupt provision of information" according to the present exemplary embodiment.

[0036] Here, in the response system **10** according to the present exemplary embodiment, when the end button **52** is selected, any one of confirmation indications in FIGS. **6** to **8** is displayed.

[0037] More specifically, in the response system **10** according to the present exemplary embodiment, when the CPU **24**A determines that the sufficiency level of information is low at the timing when the end button **52** is selected, a first confirmation indication **62** illustrated in FIG. **6** is displayed. As illustrated in FIG. **6**, a first work details notification **62**M, a leave button **70**, and a continue button **72** are displayed in the first confirmation indication **62**.

[0038] In the first work details notification **62**M illustrated in FIG. **6**, it is described that the cause of the failure occurring in the printer **12** has failed to be narrowed down based on the information provided from the user UR, and that the repair person RR cannot be dispatched. Furthermore, it is described in the first work details notification **62**M that the user UR needs to make a phone call to a responder or the like in a call center to explain the failure occurring in the printer **12**. The user UR making the phone call to the call center is an example of the "work details of the user UR" according to the present exemplary embodiment.

[0039] In the present exemplary embodiment, in a case where the user UR selects the leave button **70** in a state where the first confirmation indication **62** is displayed, the response system **10** ends. Thereafter, the user UR will make a phone call to the call center, to explain the failure of the printer **12**, for example. On the other hand, in a case where the user UR selects the continue button **72** in a state where the first confirmation indication **62** is displayed, the first confirmation indication **62** is erased, and the conversation with the chatbot CB is resumed as illustrated in FIGS. **3** to **5**. The selection of the leave button **70** in a state where the first confirmation indication **62** is displayed to leave the conversation with the chatbot CB is an example of "a case where the provision of the information is interrupted and a transition is made to another way of solution" according to the present exemplary embodiment.

[0040] Furthermore, in the response system **10** according to the present exemplary embodiment, a second confirmation indication **64** illustrated in FIG. **7** is displayed when the CPU **24**A determines that the sufficiency level of information is medium at the timing when the end button **52** is selected. As illustrated in FIG. **7**, a second work details notification **64**M, the leave button **70**, and the continue button **72** are displayed in the second confirmation indication **64**.

[0041] The second work details notification **64**M illustrated in FIG. **7** indicates that the cause of the failure occurring in the printer **12** has not been satisfactorily narrowed down based on the information provided from the user UR, and that an inquiry for confirmation may be made to the user UR. In other words, it is described in the second work details notification **64**M that the user UR may further respond to the inquiry. The user UR further responding to the inquiry is another example of the "work details of the user UR" according to the present exemplary embodiment.

[0042] In the present exemplary embodiment, in a case where the user UR selects the leave button **70** in a state where the second confirmation indication **64** is displayed, the response system **10** ends. However, the user UR may receive a phone call later on from the call center, for example, and explain the failure of the printer **12**. On the other hand, in a case where the user UR selects the continue button **72** in a state where the second confirmation indication **64** is displayed, the second confirmation indication **64** is erased, and the conversation with the chatbot CB is resumed as illustrated in FIGS. **3** to **5**.

[0043] Furthermore, in the response system **10** according to the present exemplary embodiment, a third confirmation indication **66** illustrated in FIG. **8** is displayed when the CPU **24**A determines that the sufficiency level of information is high at the timing when the end button **52** is selected. As illustrated in FIG. **8**, a third work details notification **66**M, the leave button **70**, and the continue button **72** are displayed in the third confirmation indication **66**.

[0044] It is described in the third work details notification **66**M illustrated in FIG. **8** that the cause of the failure occurring in the printer **12** has been satisfactorily narrowed down based on the information provided from the user UR, and therefore, the repair person RR is to be dispatched. In other words, it is described in the third work details notification **66**M that the user UR does not need to further provide information on the failure occurring in the printer **12**.

[0045] In the present exemplary embodiment, when the user UR selects the leave button **70** in a state where the third confirmation indication **66** is displayed, the response system **10** ends. Thereafter, a screen for inputting the date and time when the repair person is to be dispatched is displayed to the user UR as an example. On the other hand, when the user UR selects the continue button **72** in a state where the third confirmation indication **66** is displayed, the third confirmation indication **66** is erased, and the conversation with the chatbot CB is resumed as illustrated in FIGS. **3** to **5**.

[0046] Next, a procedure for receiving the details of the failure from the user UR, which is executed by the CPU **24**A of the response system **10** according to the present exemplary embodiment, will be described.

(Procedure for Receiving Details of Failure From User UR)

[0047] The program **25** of the response system **10** executed by the CPU **24**A includes a procedure illustrated in FIG. **9**. Although not elaborated in FIG. **9**, the CPU **24**A assumes, as the initial value, the sufficiency level of information to be "low" at the time when the procedure of receiving the details of the failure from the user UR is started.

[0048] First of all, in step S**102**, the CPU **24**A receives an input from the user UR. More specifically, in step S**102**, the CPU **24**A waits until the end button **52** is selected or new information is provided. Next, when the end button **52** is selected or new information is provided, the CPU **24**A proceeds to step S**104**.

[0049] Next, in step S**104**, the CPU **24**A determines whether the end button **52** has been selected. Upon determining NO in step S**104**, in other words, when new information is provided from the user UR, the CPU **24**A proceeds to step S**106**. On the other hand, upon determining YES in step S**104**, in other words, when the user UR selects the end button **52**, the CPU **24**A proceeds to step S**120**.

[0050] Next, in step S**106**, the CPU **24**A analyzes the information provided from the user UR and estimates the cause of the failure occurring in the printer **12**. More specifically, the CPU **24**A estimates the cause of the failure occurring in the printer **12** based on the information provided from the user UR and input to the chat display area **40**. Then, the CPU **24**A proceeds to step S**108**.

[0051] Next, the CPU **24**A determines the number of estimated causes of the failure in step S**108**. More specifically, in step S**108**, the CPU **24**A determines whether more than one (two or more) cause of the failure has been estimated in step S**106**. The case where more than one cause of the failure has been estimated is an example of a "case where the sufficiency level of information does not satisfy a predetermined threshold" according to the present exemplary embodiment. Upon determining NO in step S**108**, the CPU **24**A proceeds to step S**110**. On the other hand, upon determining YES in step S**108**, the CPU **24**A proceeds to step S**112**.

[0052] Next, in step S**110**, the CPU **24**A determines how many times the same result has been obtained in succession for the cause of the failure occurring in the printer **12** estimated in step S**106**. More specifically, the CPU **24**A determines whether all the estimation results are the same in the latest predetermined number of times of the number of times step S**106** has been executed. Note that a case where all the estimation results are the same in the latest predetermined number of times is an example of a "case where a sufficiency level of the information exceeds a predetermined threshold" according to the present exemplary embodiment. Upon determining YES in step S**110**, the CPU **24**A proceeds to step S**116**. On the other hand, upon determining NO in step S**110**, in other words, when different estimation results are obtained, or when step S**106** has not been executed for a predetermined number of times yet, the CPU **24**A proceeds to step S**114**.

[0053] Next, upon proceeding to step S**112**, the CPU **24**A determines that the sufficiency level of information is "low". Then, the CPU **24**A proceeds to step S**117**.

[0054] Upon proceeding to step S**114**, the CPU **24**A determines that the sufficiency level of information is "medium". Then, the CPU **24**A proceeds to step S**117**.

[0055] Upon proceeding to step S**116**, the CPU **24**A determines that the sufficiency level of information is "high". Then, the CPU **24**A proceeds to step S**117**.

[0056] Next, in step S**117**, the CPU **24**A updates the display of the current value **56** of the work amount meter **54** in accordance with the sufficiency level of information updated in step S**112**, step S**114**, or step S**116**. More specifically, upon determining the sufficiency level of information to be "low", the CPU **24**A displays the work amount meter **54** the entirety of which is filled as illustrated in FIG. **3**. Upon determining the sufficiency level of information to be "medium", the CPU **24**A displays the work amount meter **54** about two thirds of which is filled as illustrated in FIG. **4**. Upon determining the sufficiency level of information to be "high", the CPU **24**A displays the work amount meter **54** about one third of which is filled as illustrated in FIG. **5** for example. Then, the CPU **24**A proceeds to step S**118**.

[0057] Next, in step S**118**, the CPU **24**A displays the next question **36** in the chat display area **40**. More specifically, in step S**118**, the CPU **24**A displays the new question **36** in the chat display area **40**, and prompts the user UR to provide new information. Then, the CPU **24**A proceeds to step S**102**.

[0058] Upon proceeding to step S**120**, the CPU **24**A displays a confirmation indication corresponding to the sufficiency level of information on the input screen **32**. More specifically, when the user UR selects the end button **52** in a state where the sufficiency level of information is determined to be "low", the CPU **24**A displays the first confirmation indication **62** illustrated in FIG. **6** as the confirmation indication for the sufficiency level of information "low". When the user UR selects the end button **52** in a state where the sufficiency level of information is determined to be "medium", the CPU **24**A displays the second confirmation indication **64** illustrated in FIG. **7** as the confirmation indication for the sufficiency level of information "medium". When the user UR selects the end button **52** in a state where the sufficiency level of information is determined to be "high", the CPU **24**A displays the third confirmation indication **66** illustrated in FIG. **8** as the confirmation indication for the sufficiency level of information "high". Then, the CPU **24**A proceeds to step S**122**.

[0059] Next, in step S**122**, the CPU **24**A determines whether the leave button **70** is selected. More specifically, the CPU **24**A determines in step S**122** whether the leave button **70** in the first confirmation indication **62**, the second confirmation indication **64**, or the third confirmation indication **66** has been selected. Then, upon determining NO in step S**122**, the CPU **24**A erases the first confirmation indication **62**, the second confirmation indication **64**, or the third confirmation indication **66**, and then proceeds to step S**102**. On the other hand, upon determining YES in step S**122**, the CPU **24**A ends the procedure for receiving the details of the failure from the user UR.

[0060] Note that in the present exemplary embodiment, the CPU **24**A displays a telephone number to the call center on the display screen of the input device **30** for the user UR when ending the procedure for receiving the details of the failure from the user UR, with the sufficiency level of information being "low". The CPU **24**A provides the information obtained based on the information provided from the user UR to the repair person RR, when the procedure for receiving the details of the failure from the user UR ends with the sufficiency level of information being "medium" or "high".

[0061] A larger amount of information on the failure having occurred in the printer **12** provided from the user UR leads to a lower risk for the repair person RR encountering an unexpected situation. In other words, the information on the failure provided from the user UR to the repair person RR may be provided until it is determined YES in step S**110** described above.

[0062] Therefore, in the response system **10** according to the present exemplary embodiment, the CPU **24**A further displays the question **36** to the user UR even after proceeding to step S**116** illustrated in FIG. **9**, meaning that the sufficiency level of information is "high". In other words, the CPU **24**A repeats steps S**102** to S**118** illustrated in FIG. **9** to further continue prompting the user UR to provide information, thereby causing calculation of the sufficiency level of information.

[0063] With the response system **10** and the program **25** according to the present exemplary embodiment, the following operations and effects are achieved.

(Operations and Effects)

[0064] The response system **10** according to the present exemplary embodiment requests the user UR who uses the printer **12** in which a failure has occurred to provide information on the failure of the printer **12**, and displays the status display area **50**. Therefore, with the response system **10** according to the present exemplary embodiment, in the response system **10** that collects information provided from the user UR who uses the printer **12** in which the failure has occurred, it is possible to prompt the user UR to continue providing information.

[0065] The response system **10** according to the present exemplary embodiment displays the work amount meter **54** indicating whether the amount of work required when the user UR selects the leave button **70** is large or small. Therefore, with the response system **10** according to the present exemplary embodiment, the provision of information from the user UR is more likely to be continued than in a case where the work details required by the user UR are directly displayed using only the status display area **50**.

[0066] The response system **10** according to the present exemplary embodiment displays, as the first confirmation indication **62**, the second confirmation indication **64**, or the third confirmation indication **66**, the work details of the user UR required when the user UR selects the leave button **70** in a state where the user UR has selected the end button **52**. Thus, with the response system **10** according to the present exemplary embodiment, the voluntary provision of information by the user UR is less likely to be interrupted than in a case where only the status display area **50** is always displayed, and thus the user UR is likely to concentrate on the provision of information.

[0067] The response system **10** according to the present exemplary embodiment continues requesting the provision of information until the user UR attempts to select the end button **52**. Therefore, with the response system **10** according to the present exemplary embodiment, the voluntary provision of information by the user UR is less likely to be interrupted than in a case where the amount of information received from the user UR is limited, and thus a larger amount of information is likely to be provided from the user UR.

[0068] The response system **10** according to the present exemplary embodiment further displays the work amount meter **54** indicating the information sufficiency level even when the sufficiency level of the information provided from the user UR exceeds the threshold. Therefore, with the response system **10** according to the present exemplary embodiment, compared with a case where the display of the current value **56** of the work amount meter **54** is stopped when the information on the failure of the printer **12** is sufficiently accumulated, it is possible to prompt the user UR to continue the provision of information.

[0069] The program **25** according to the present exemplary embodiment causes a processor to execute requesting the user UR who uses the printer **12** in which the failure has occurred to provide information on the failure of the printer **12**, and displaying the first confirmation indication **62** when the information sufficiency level included in the information provided from the user UR is lower than the predetermined threshold. Therefore, with the program **25** according to the present exemplary embodiment, in a program for collecting information provided from the user UR who uses the printer **12** in which the failure has occurred, it is possible to prompt the user UR to continue providing information.

(Modifications)

[0070] In the above description, the CPU **24**A provides the user UR with the sufficiency level of information by displaying the work amount meter **54**. However, the response system **10** according to the present exemplary embodiment is not limited thereto as long as it is possible to provide the user UR with the current value **56** of the sufficiency level of information. For example, instead of the method of displaying the progress bar as the work amount meter **54**, the information sufficiency level may be displayed by a numerical value, or may be indicated by the size, shape, position, color, or the like of an icon or the like displayed in the status display area **50**. Further, instead of the display using the status display area **50**, the information sufficiency level may be provided to the user UR as auditory information such as sound emitted.

[0071] Furthermore, in the above description, the CPU **24**A includes, in the first confirmation indication **62**, the description indicating that the user UR makes a phone call to the call center, but the response system **10** according to the present exemplary embodiment is not limited thereto. For example, as the first work confirmation indication, only a message indicating that the repair person RR cannot be dispatched may be displayed, and the description indicating that the user UR also needs to make a phone call to the call center may be omitted.

[0072] Furthermore, in the above description, the CPU **24**A displays the first work details notification **62**M when the user UR selects the end button **52**, but the response system **10** according to the present exemplary embodiment is not limited thereto. For example, a mode may be adopted in which the CPU **24**A constantly displays the first work details notification **62**M in the status display area **50**.

[0073] In the above description, the CPU **24**A keeps asking the user UR questions until the user UR selects the end button **52**. However, the response system **10** according to the present exemplary embodiment is not limited thereto. For example, when the sufficiency level of information is determined to be "high", the CPU **24**A may stop asking the user UR additional questions.

[0074] Furthermore, while the CPU **24**A continues displaying the work amount meter **54** even after determining that the sufficiency level of information is "high" in the description above, the response system **10** according to the present exemplary embodiment is not limited thereto. For example, when the CPU **24**A determines that the sufficiency level of information is "high", the work amount meter **54** displayed may be erased.

[0075] Even with these modifications, it is possible to obtain the same operations and effects as those described above.

[0076] While the exemplary embodiments of the present disclosure have been described with reference to the accompanying drawings, it is apparent that those skilled in the art to which the present disclosure pertains can conceive various modifications and applications within the scope of the technical idea described in the claims, and it is to be understood that these naturally belong to the technical scope of the present disclosure.

Appendix

(((1)))

[0077] An information processing system comprising a processor configured to, when a solution request for a failure of a device in which the failure has occurred is received from a user who uses the device, request the user to provide information related to solution for the failure, and display, to the user, an indication related to an amount of work required from the user when a transition is made to another way of solution for the failure.

(((2)))

[0078] The information processing system according to (((1))), wherein the indication includes whether an amount of work required when the user interrupts provision of information is large or small.

(((3)))

[0079] The information processing system according to (((1))) or (((2))), wherein in a case where the user attempts to interrupt provision of information, the processor is configured to display an amount of work for the user when a transition is made to another way of solution.

(((4)))

[0080] The information processing system according to any one of (((1)) to (((3))), wherein the processor is configured to continue requesting for provision of information until the user attempts to interrupt provision of information.

(((5)))

[0081] The information processing system according to (((4))), wherein the processor is configured to display the indication also in a case where a sufficiency level of the information on the failure based on provision of the information from the user exceeds a predetermined threshold.

(((6)))

[0082] A program causing a processor to execute a process comprising:

[0083] requesting, when a solution request for a failure of a device in which the failure has occurred is received from a user who uses the device, the user to provide information related to solution for the failure; and

[0084] displaying, to the user, an indication related to an amount of work required from the user when a transition is made to another way of solution for the failure.

What is claimed is:

1. An information processing system comprising:

a processor configured to, when a solution request for a failure of a device in which the failure has occurred is received from a user who uses the device, request the user to provide information related to solution for the failure, and display, to the user, an indication related to an amount of work required from the user when a transition is made to another way of solution for the failure.

2. The information processing system according to claim **1**, wherein the indication includes whether an amount of work required when the user interrupts provision of information is large or small.

3. The information processing system according to claim **1**, wherein in a case where the user attempts to interrupt provision of information, the processor is configured to display work details for the user when a transition is made to another way of solution.

4. The information processing system according to claim **1**, wherein the processor is configured to continue requesting for provision of information until the user attempts to interrupt provision of information.

5. The information processing system according to claim **2**, wherein the processor is configured to continue requesting for provision of information until the user attempts to interrupt provision of information.

6. The information processing system according to claim **3**, wherein the processor is configured to continue requesting for provision of information until the user attempts to interrupt provision of information.

7. The information processing system according to claim **4**, wherein the processor is configured to display the indication also in a case where a sufficiency level of the information on the failure based on provision of the information from the user exceeds a predetermined threshold.

8. The information processing system according to claim **5**, wherein the processor is configured to display the indication also in a case where a sufficiency level of the information on the failure based on provision of the information from the user exceeds a predetermined threshold.

9. The information processing system according to claim **6**, wherein the processor is configured to display the indication also in a case where a sufficiency level of the information on the failure based on provision of the information from the user exceeds a predetermined threshold.

10. A non-transitory computer readable medium storing a program causing a processor to execute a process comprising:

requesting, when a solution request for a failure of a device in which the failure has occurred is received from a user who uses the device, the user to provide information related to solution for the failure; and

displaying, to the user, an indication related to work details required by the user when a transition is made to another way of solution for the failure.

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